

Problem

The *PSpice* part name for a current-controlled voltage source is:

- (a) EX
- (b) FX
- (c) HX
- (d) GX

Step-by-step solution

Step 1 of 5

Refer to Figure 10.49 in the textbook.

Draw the circuit as shown in Figure 1.

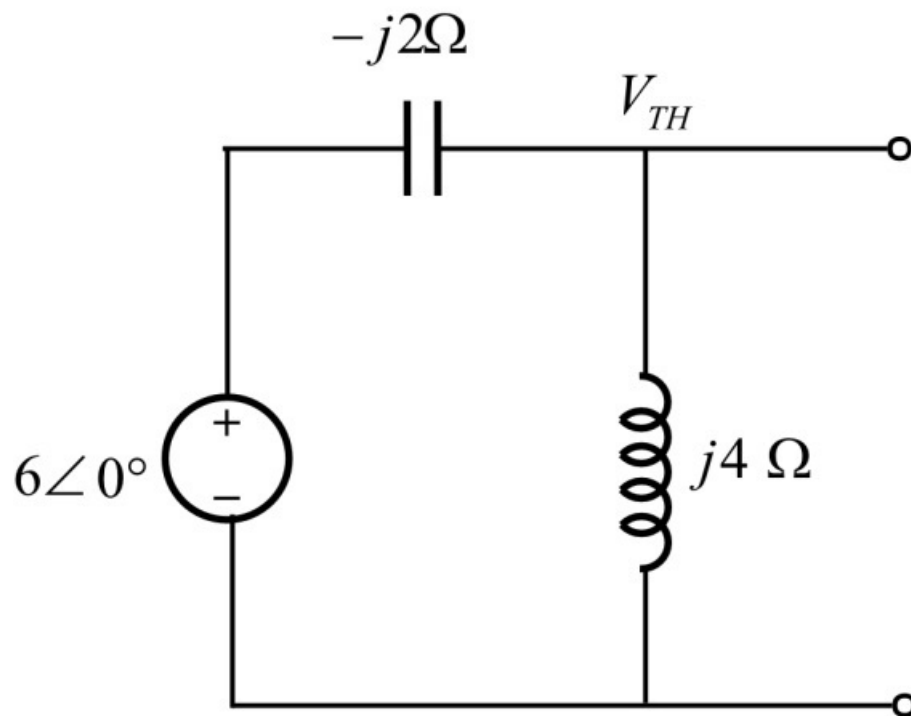


Figure 1

Step 2 of 5

Determine the Thevenins equivalent circuit.

Apply Kirchhoff's current law at node V_{TH} .

$$\frac{V_{TH} - 6\angle 0^\circ}{-j2} + \frac{V_{TH}}{j4} = 0$$

$$\frac{V_{TH}}{-j2} - \frac{6\angle 0^\circ}{-j2} + \frac{V_{TH}}{j4} = 0$$

$$\frac{jV_{TH}}{2} - j\frac{6\angle 0^\circ}{2} - \frac{jV_{TH}}{4} = 0$$

$$\frac{jV_{TH}}{4} = j\frac{6\angle 0^\circ}{2}$$

$$\frac{V_{TH}}{4} = \frac{6\angle 0^\circ}{2}$$

$$V_{TH} = 12\angle 0^\circ$$

Step 3 of 5

Determine the equivalent Thevenins resistance R_{TH} .

Draw the circuit as shown in Figure 2 to determine the equivalent resistance.

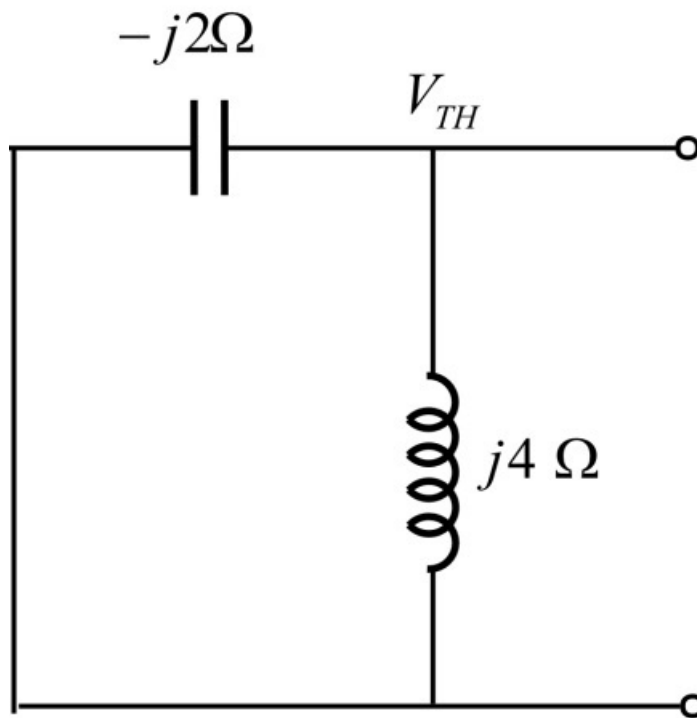


Figure 2

Step 4 of 5

Consider $-j2$ and $j4$ are in parallel combination.

$$R_{TH} = \frac{(-j2)(j4)}{j4 - j2}$$

$$= \frac{8}{j2}$$

$$= -j4\Omega$$

Step 5 of 5

Determine the Norton current I_N .

$$\begin{aligned} I_N &= \frac{V_{TH}}{R_{TH}} \\ &= \frac{12\angle 0^\circ}{-j4} \\ &= j3 \text{ A} \\ &= 3\angle 90^\circ \text{ A} \end{aligned}$$

Therefore, the Norton current is $3\angle 90^\circ$.

Therefore, the correct option is **d**.